Abhishek S

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B.Tech '26 · IISc · 5 ICRA/RSS workshop papers

Work Experience

- **Indian Institute of Science (Project Associate)** *Bengaluru, India*
  Advisor: *Dr. Jishnu Keshavan, Assistant Professor, IISc* *September 2026 – Present*
  Developing a cross-embodiment generalist control policy for multirotors across quadrotor and hexarotor configurations.
- **Indian Institute of Science (Control and Optimization Intern)** *Bengaluru, India*
  Advisor: *Dr. Pavan Tallapragada, Associate Professor, IISc* *August 2025 – July 2026*
  Designed a real-time multi-robot controller and coordinator to bridge planning and safe execution, using quadrant-token mutual exclusion, arrival-time arbitration, and wait-for-graph cycle detection; conflict-free, scalable to 10 robots, validated across 125+ OptiTrack experiments, maintaining spatio-temporal accuracy with 0.25 s schedule error and 0.12 s inter-robot wait.

Published Research

- **Teaching Robots to Say 'I Don't Know': SENTINEL for Uncertainty SLAM (ICRA 2026@UOWR)** [arxiv]
  Engineered a **training-free** reliability layer flagging silent LiDAR failures on sub-$300 robots, fusing scan geometry with LiDAR/RGB-D depth consistency to reject corrupted scans via reliability scores, validated across **5 failure modes**.
- **CADENCE: Predicting Realized MAPF Execution Time Beyond Sum of Costs (ICRA 2026@MARS)** [arxiv]
  Built a **480-trial robotic hardware corpus** (7 robots) to predict completion time, showing **primitive motion burden** cuts held-out error by **48.6–59.8% MAE** over Sum-of-Costs via scenario-held-out ridge and mixed-effects models.
- **MineXplore: A RL Exploration Benchmark for GNSS-Denied Environment (ICRA 2026@Xplore)** [arxiv]
  Addressing the absent mining-RL benchmark, built an open-source MuJoCo simulation of a **real Chilean copper mine** via a contour-to-MJCF pipeline, achieving **95.38% geometric IoU** and validating PPO policies at **88.9% coverage**.
- **Subterra: A Validated Benchmark for GNSS-Denied Tunnel Navigation (RSS 2026@WPCIS)** [link]
  Lacking a real-geometry tunnel-RL testbed, rebuilt a **351 m** metro tunnel from LiDAR scans as a MuJoCo POMDP, validated at **97.1% fidelity** (2.46 cm surface error) across 199 metrics, where a PPO agent reached **90.02% coverage**.
- **Beyond SoC: Plan-Space Sampling for Execution-Time Optimization in MAPF (RSS 2026@SBO, Oral)** [link]
  Developed **EA-LNS**, an anytime plan-space MAPF optimizer that repairs agent neighborhoods using an analytical ADG completion-time estimate and evaluates accepted plans in SMART; reduced average execution time by **19–53%** versus equal-budget MAPF-LNS2 across **345 paired instances**, matching rotation-aware optimal planning on open maps and scaling beyond it in congested warehouses.

Projects

‡ *IISc* † *PES University*

- **ETC-RL: Lyapunov-Safe Event-Triggering Policies**‡ *August 2025 – November 2025*
  Cast event-triggered control of a linear system as a Markov decision process and synthesised state-dependent triggering policies via safe RL spanning tabular and deep methods (Q-learning, SAC) maximising a discounted inter-event-time objective subject to Lyapunov-derived bounds on the triggering threshold that preserve closed-loop stability; proving the inter-event time exhibits intrinsic jump discontinuities in the triggering parameter β.
- **EdgeLIO: LiDAR–Inertial Perception Stack for Edge Mobile Robots**† **[github]** *January 2025 – May 2025*
  Engineered an onboard LiDAR–inertial perception stack (**FAST-LIO2**, **Livox MID360**, source-built **ROS 2 Humble** on Ubuntu 20.04) for real-time state estimation and 3D mapping on a custom Jetson NX mobile robot. Integrated LiDAR/IMU sensing and Arduino serial control, sustaining 10 Hz mapping over a 15-minute physical indoor run at ∼22% CPU with 15 ms/scan latency and 0.13 m ATE RMSE, enabling downstream autonomous navigation without offboard compute.
- **SceneForge: Interactive 3D Object Transform Editor**† **[github] [live demo]** *August 2024 – December 2024*
  Built an in-browser 3D object-transform editor (**React Three Fiber**, **Three.js**) with draggable numeric inputs for precise position/rotation/scale in real-time two-way sync with an in-viewport gizmo, plus a movable properties panel with per-axis increment locking.

Education

PES University August 2022 – July 2026
Bachelor of Technology, Computer Science and Engineering *CGPA: 7.7 / 10*

Skills & Tools

Research Areas: Robot Foundation Models, Cross-Embodiment Learning, Motion Planning & Control, Reliability-Aware Robot Perception & SLAM, Reinforcement Learning for Robotics, Real-to-Sim Reconstruction & Fidelity Validation
Tools: *Programming:* Python, C++; *Robotics & Simulation:* ROS 1/2, MuJoCo, Gazebo, MATLAB/Simulink, FAST-LIO2, Open3D; *Learning:* PyTorch, TensorFlow, Stable-Baselines3, RLlib; *Hardware:* NVIDIA Jetson Xavier NX, Arduino, Livox MID360, OptiTrack